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INSTITUTE OF ENGINEERING AND TECHNOLOGY
Approved by AICTE & Affiliated to APJ Abdul Kalam Technological University
(Owned by Adi Sankara Trust)
DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE)

AIL 411-DEEP LEARNING LAB

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AP,CSE(AI),ASIET,KALADY

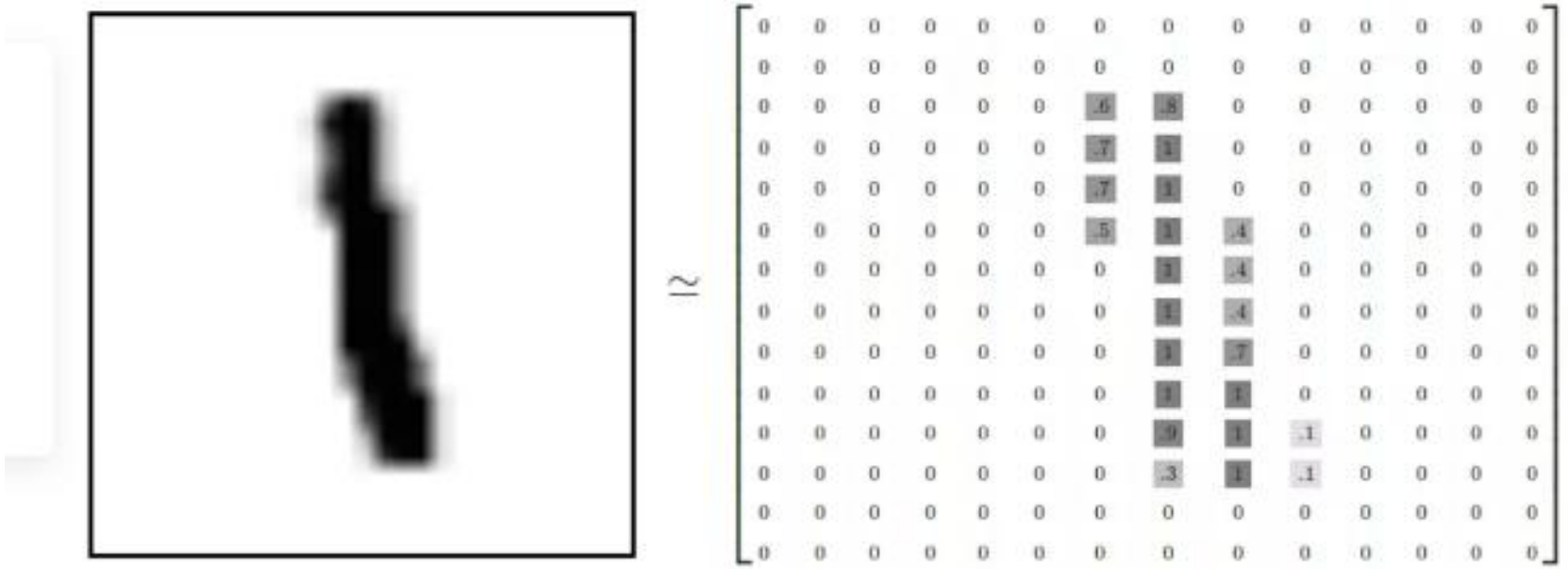
EXPERIMENT 5

Digit classification using CNN architecture for MNIST dataset

MNIST Dataset

- **MNIST Database**:
  - Stands for Modified National Institute of Standards and Technology database.
  - Large database of handwritten digits.
  - Commonly used for training image processing systems and machine learning models.
- **Image Normalization**:
  - Black and white images normalized to fit into a 28x28 pixel bounding box.
  - Anti-aliased to introduce grayscale levels.
- **Dataset Composition**:
  - Contains 60,000 training images and 10,000 testing images.

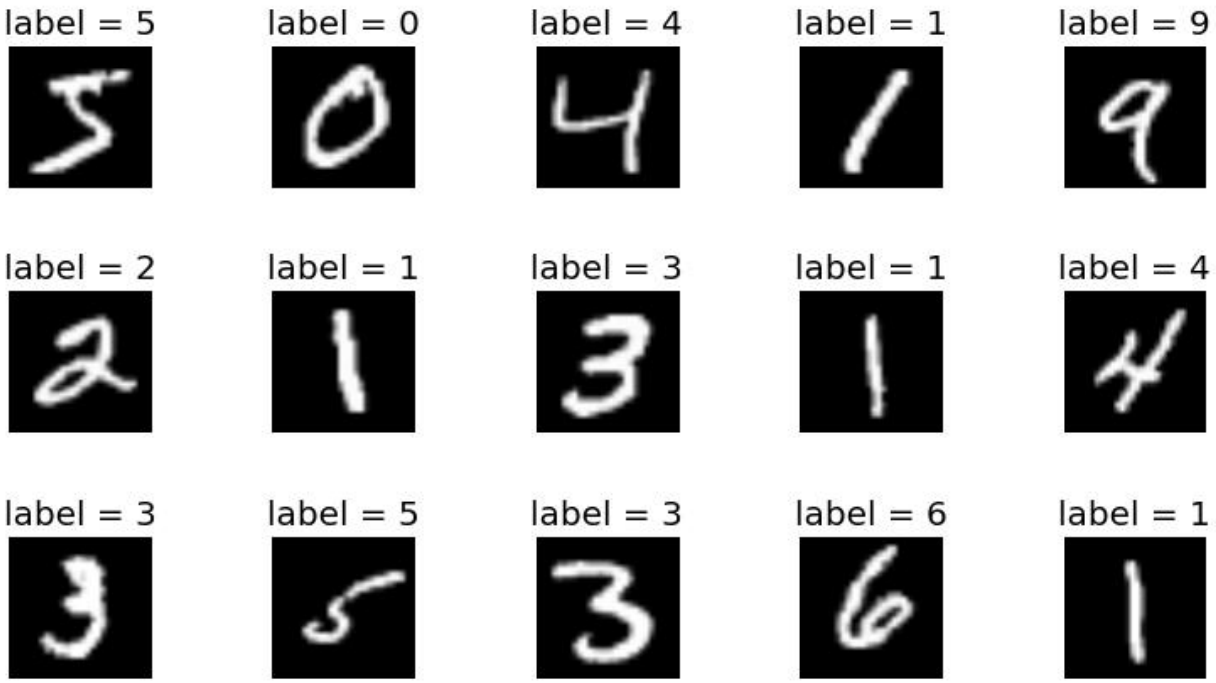
MNIST Dataset



28x28 pixel images of hand-written digits

Every image can be thought of as an **array** of numbers between **0** and **1** describing how dark each pixel is (intensity of pixel)

MNIST Dataset



Every MNIST data point has **two parts**:

1. **Image** of a handwritten digit
2. Corresponding **label** (number between **0** and **9**) representing the digit drawn in the image.

Reference